Slide 1: Killing Agent Roulette

The Probabilistic Fallacy of Agentic DB Execution

- Deploying autonomous LLM agents with direct database access is **architectural suicide**.
- Relying on probabilistic prompt templates to enforce catalog permissions invites rogue execution.
- LLM agentic tool-calls must be governed by deterministic, zero-trust gatekeepers.

Slide 2: The MCP Security Gap

How Semantic Jailbreaks Bypass Gateways

- Model Context Protocol (MCP) payloads are generated dynamically by probabilistic models.
- Upstream prompt injections alter SQL parameters or request paths (e.g., `drop_table`, `../../tenant_id`).
- Without inline interdiction, the target database receives a validly formatted but **highly malicious payload**.

Slide 3: OBO Token Interdiction

The Natoma Edge Architecture

- We deploy a localized **Natoma Ingress Interdiction Proxy** at the edge boundary.
- Intercepted bearer tokens undergo asymmetric **RS256 JWT validation** mimicking the Unity Catalog endpoint.
- Validated payloads are matched against a strict Metadata Containment Map before outbound routing.

Slide 4: Deterministic Containment

Proof of Isolation

- Verified zero-bypass boundaries using our integration test harness:
    - **Valid Read Queries**: Executed and routed cleanly via Oracle Egress nodes.
    - **SQL Injection Attacks**: Blocked instantly returning strict JSON-RPC 403 errors.
    - **Cross-Tenant Directory Traversal**: Prevented at the proxy containment layer.

Slide 5: Secure Your Enclaves

Get the Sovereign Architecture Code

- Stop trusting generative intent; enforce **Zero-Trust, Zero-Hop** gateway boundaries.
- Read the full architectural whitepaper containing FastAPI and OBO validator configurations.
- Access the production-ready code repository:
  - **Link to GitHub & Full Article in Comments Below!**